

7 When antagonistic muscles contract, they move skeletal bones.

(a) (i) Energy is needed when a muscle contracts.

Which row in the table shows the energy use in muscle contraction?

(1)

		Energy in muscle contraction is used to		
		break the myosin–actin cross bridge	remove calcium ions from the sarcoplasm	release ADP and Pi from ATP
☒	A	✓	✓	✓
☒	B	✓	✓	X
☒	C	X	X	✓
☒	D	X	X	X

(ii) Describe the sliding filament theory of muscle contraction.

(4)

- (b) Two types of ostrich, the South African Black and the Zimbabwean Blue, are farmed in South Africa.

Ostrich muscle is a source of dietary protein.

Scientists compared the mean masses of three muscles from the ostriches.

Muscle	Mean mass of muscle and standard deviation / kg	
	South African Black (n = 34)	Zimbabwean Blue (n = 2)
T	0.69 ± 0.10	0.88 ± 0.11
G	0.59 ± 0.08	0.74 ± 0.03
F	0.40 ± 0.06	0.50 ± 0.01

- (i) State **two** conclusions that can be drawn about the mean mass of muscles shown in the table.

(2)

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- (ii) Comment on the validity of conclusions drawn from the data for these two groups of ostrich.

(2)

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- (iii) Calculate the percentage difference in the mean mass of muscle G of the South African Black and the Zimbabwean Blue ostriches.

Use the equation: percentage difference between a and b = $\frac{a - b}{\left(\frac{a + b}{2}\right)} \times 100$

Give your answer to **two** decimal places.

(2)

Answer%

- (c) The hypothalamus is involved in maintaining a steady internal temperature in the ostrich.

Describe the role of the hypothalamus in maintaining a steady internal temperature.

(3)

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(Total for Question 7 = 14 marks)